

10:41

4G 82

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✓ **Correct** 1/1 Points

9. Which statement BEST explains the purpose of caching in REST? \*

- ☐ It ensures every request reaches the database
- ☒ It reduces repeated server processing and improves performance ✓
- ☐ It stores session data for authentication
- ☐ It prevents unauthorized access
- ☐ It guarantees real-time data consistency

✗ **Incorrect** 0/1 Points

10. Why is the Uniform Interface constraint critical to REST architecture? \*

- ☐ It ensures faster database queries
- ☐ It enforces the use of JSON only
- ☐ It standardizes communication and enables



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✓ **Correct** 1/1 Points

3. Which of the following is an optional constraint in REST architecture? \*

- ☐ Uniform Interface
- ☐ Stateless
- ☐ Cacheable
- ☒ Code on Demand
- ☐ Layered System



✓ **Correct** 1/1 Points

4. What is the key feature that differentiates a Level-3 API in the Richardson Maturity Model? \*

- ☐ Use of HTTP verbs
- ☐ Use of JSON format
- ☒ Implementation of HATEOAS



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## IT3030 PAF Lecture 05 Quiz

REST APIs

Points: 7/10

✓ **Correct** 1/1 Points

1. What is the primary goal of the Client-Server constraint in REST architecture? \*

- ☐ To improve the server's response time.
- ☒ To maintain a clear separation between client and server for better scalability and portability. ✓
- ☐ To allow clients to store session data for faster processing.
- ☐ To ensure APIs support only HTTP methods
- ☐ To eliminate the need for middleware

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✓ **Correct** 1/1 Points

6. Which scenario BEST violates the Stateless constraint? \*

- ☐ Each request includes an authentication token
- ☐ The client sends complete request headers each time
- ☐ Responses include cache-control headers
- ☐ The client uses HTTPS for secure communication
- ☒ The server stores user session state between requests

✗ **Incorrect** 0/1 Points

7. Which response best demonstrates a RESTful uniform interface with HATEOAS? \*

- ☐ { "id": 10, "name": "Nimal" }
- ☐ { "status": "success" }



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✗ **Incorrect** 0/1 Points

8. Which of the following designs is MOST RESTful? \*

- ☒ /getAllUsers
- ☐ /deleteUser?id=10
- ☐ /users/10/delete
- ☐ DELETE /users/10
- ☐ /users?action=delete&id=10



✓ **Correct** 1/1 Points

9. Which statement BEST explains the purpose of caching in REST? \*

- ☐ It ensures every request reaches the database
- ☒ It reduces repeated server processing and improves performance
- ☐ It stores session data for authentication





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✗ **Incorrect** 0/1 Points

10. Why is the Uniform Interface constraint critical to REST architecture? \*

- ☐ It ensures faster database queries
- ☐ It enforces the use of JSON only
- ☒ It standardizes communication and enables independent evolution of clients and servers ✓
- ☐ It prevents network failures
- ☐ It eliminates the need for authentication

Keep the information with you by saving your response.

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✗ **Incorrect** 0/1 Points

7. Which response best demonstrates a RESTful uniform interface with HATEOAS? \*

- ☐ { "id": 10, "name": "Nimal" }
- ☐ { "status": "success" }
- ☐ { "id": 10, "name": "Nimal", "orders": "/orders/10" } ✓
- ☐ { "message": "User fetched" }
- ☒ { "user": 10 }

✗ **Incorrect** 0/1 Points

8. Which of the following designs is MOST RESTful? \*

- ☒ /getAllUsers
- ☐ /deleteUser?id=10



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Model? \*

- ☐ Use of HTTP verbs
- ☐ Use of JSON format
- ☒ Implementation of HATEOAS ✓
- ☐ Multiple endpoints per resource
- ☐ Stateless communication

✓ **Correct** 1/1 Points

5. In REST, what does the term 'Resource' represent? \*

- ☐ The server's database schema.
- ☒ Any content accessible via a URI ✓
- ☐ A client-side function
- ☐ A web interface component
- ☐ A stored procedure in the database





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✓ **Correct** 1/1 Points

2. Which of the following best describes the 'Stateless' constraint in REST APIs? \*

- ☐ The server stores session data for faster processing
- ☐ The client and server maintain a persistent connection
- ☒ Each request contains all required information, and the server stores no client context ✓
- ☐ The server caches all responses automatically
- ☐ The client stores all application logic

✓ **Correct** 1/1 Points

3. Which of the following is an optional constraint in REST architecture? \*

- ☐ Uniform Interface
- ☐ Stateless



10:36

4G 78

- ☐ Backend services are deployed independently

✗ **Incorrect** 0/1 Points

8. How does the “Big Ball of Mud” architectural anti-pattern emerge in web applications, and what strategies can be used to prevent it? \*

- ☐ It results from excessive code reuse, prevented by writing more monolithic applications.
- ☐ It is caused by excessive use of microservices, prevented by migrating back to a monolithic architecture.
- ☐ It occurs due to unstructured code and lack of modularity, can be prevented using layered architecture principles. ✓
- ☐ It is an unavoidable issue in all software systems with complex logic.

✓ **Correct** 1/1 Points

9. Considering the evolution of frontend



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10:36

4G 78

- ☐ It requires the entire application to be rewritten in one language.
- ☐ It has no infrastructure costs.

✗ **Incorrect** 0/1 Points

7. Which situation most clearly indicates that a system is drifting toward a “Big Ball of Mud” architecture? \*

- ☐ Independent services communicate using REST APIs
- ☐ UI components are reused across multiple views
- ☐ Business logic is scattered across frontend, backend, and database layers ✓
- ☒ Backend services are deployed independently

✗ **Incorrect** 0/1 Points

8. How does the “Big Ball of Mud”



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✗ **Incorrect** 0/1 Points

3. Why does Three-Tier Architecture improve security compared to Two-Tier Architecture? \*

- ☐ Users can directly access the database layer
- ☐ Business logic and data are isolated from the client ✓
- ☐ Frontend frameworks enforce encryption automatically
- ☒ APIs eliminate the need for authentication

✓ **Correct** 1/1 Points

4. In a three-tier web application, which responsibility should NEVER belong to the Presentation Layer? \*

- ☐ Handling user input and UI rendering
- ☐ Managing session state and form validation



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Application.

✗ **Incorrect** 0/1 Points

2. What is a key advantage of Three-Tier Architecture over Two-Tier Architecture? \*

- ☐ It eliminates the need for a backend.
- ☐ It allows independent development and scaling of layers. ✓
- ☐ It does not require database interaction.
- ☒ It simplifies frontend development but makes backend development more complex.

✗ **Incorrect** 0/1 Points

3. Why does Three-Tier Architecture improve security compared to Two-Tier Architecture? \*

- ☐ Users can directly access the database layer

Business logic and data are isolated from



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✓ **Correct** 1/1 Points

4. In a three-tier web application, which responsibility should NEVER belong to the Presentation Layer? \*

- ☐ Handling user input and UI rendering
- ☐ Managing session state and form validation
- ☒ Implementing core business rules and workflows ✓
- ☐ Displaying data received from the backend

✓ **Correct** 1/1 Points

5. What is a major drawback of Monolithic Architecture? \*

- ☐ It does not support backend processing.
- ☐ It allows independent scaling of components.
- ☒ A small change requires redeploying the entire application. ✓



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4G 78

## IT3030 PAF Lecture 04 Quiz

Web Application architecture - an overview

Points: 5/10

✓ **Correct** 1/1 Points

1. What is the primary difference between a Website and a Web Application? \*

- ☐ A Website is interactive, whereas a Web Application is static.
- ☐ A Website requires a backend, while a Web Application does not.
- ☒ A Website provides static content, while a Web Application is designed for user interaction. ✓
- ☐ A Website is always more complex than a Web Application.



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- ☒ and testing of UI components, making large-scale applications easier to maintain. ✓

✗ **Incorrect** 0/1 Points

10. A development team wants to update one backend feature without redeploying the entire application. Which architectural style best supports this requirement? \*

- ☒ Two-Tier Architecture
- ☐ Monolithic Architecture
- ☐ Microservices Architecture ✓
- ☐ Big Ball of Mud Architecture

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- ☐ It is an unavoidable issue in all software systems with complex logic.

✓ **Correct** 1/1 Points

9. Considering the evolution of frontend technologies, how does Component-Based Architecture improve scalability compared to traditional MVC architectures? \*

- ☐ It eliminates the need for a backend by embedding business logic within UI components.
- ☐ It increases server load by requiring server-side rendering for all components.
- ☐ It reduces performance by adding additional abstraction layers.
- ☒ It allows independent development, reuse, and testing of UI components, making large-scale applications easier to maintain. ✓

✗ **Incorrect** 0/1 Points

10. A development team wants to update one



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✓ **Correct** 1/1 Points

5. What is a major drawback of Monolithic Architecture? \*

- ☐ It does not support backend processing.
- ☐ It allows independent scaling of components.
- ☒ A small change requires redeploying the entire application. ✓
- ☐ It is always more secure than Microservices.

✓ **Correct** 1/1 Points

6. What is a major challenge of Microservices Architecture? \*

- ☒ It is difficult to debug due to high global complexity. ✓
- ☐ It does not support independent scaling.
- ☐ It requires the entire application to be rewritten in one language.



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~ Ashfaq Ahamed

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9. A critical security vulnerability is discovered in the production system. According to best practices, from which branch should a hotfix branch be created? \*

- ☐ Develop branch
- ☐ Feature branch
- ☐ Release branch
- ☒ Main branch
- ☐ Any branch containing the bug

✓ **Correct** 1/1 Points

10. After completing a hotfix and merging it into the main branch, what is the NEXT best practice step? \*

- ☐ Leave the hotfix branch for future fixes
- ☐ Rebase the main branch onto the hotfix branch

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- ☐ Reducing the number of commits in the repository
- ☒ Isolating experimental changes from production ready code ✓
- ☐ Replacing the need for pull requests
- ☐ Preventing merge conflicts entirely

✓ **Correct** 1/1 Points

2. In a newly initialized Git repository using modern Git defaults, which branch is created automatically? \*

- ☐ develop
- ☐ feature
- ☐ hotfix
- ☒ master or main ✓
- ☐ staging





10:04

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## IT3030 PAF Lecture 03 Quiz

Version Controlling and Workflows with Git - II

Points: 12/12

✓ **Correct** 1/1 Points

1. A developer creates a new branch to implement a feature while ensuring the main branch remains stable. Which principle of Git branching does this practice demonstrate? \*

- ☐ Permanently separating development environments
- ☐ Reducing the number of commits in the repository
- ☒ Isolating experimental changes from production ready code ✓
- ☐ Replacing the need for pull requests

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8. Why is GitFlow often considered unsuitable for small student projects or small teams? \*

- ☐ It does not support feature branches
- ☒ It enforces a complex branching model that may be unnecessary ✓
- ☐ It prevents parallel development
- ☐ It cannot be used with GitHub
- ☐ It removes the need for code reviews

✓ **Correct** 1/1 Points

9. A critical security vulnerability is discovered in the production system. According to best practices, from which branch should a hotfix branch be created? \*

- ☐ Develop branch
- ☐ Feature branch
- ☐ Release branch



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6. A Pull Request (PR) is created from a feature branch to the main branch. Which is the PRIMARY reason for this step in GitHub Flow? \*

- ☐ To archive the feature branch before deletion
- ☐ To allow automated deletion of commits
- ☐ To convert the feature branch into a hotfix branch
- ☒ To enable review and validation before integrating changes into main ✓
- ☐ To prevent further commits on the feature branch

✓ **Correct** 1/1 Points

7. Which practice MOST effectively reduces merge conflicts when working with feature branches? \*

- ☐ Keeping feature branches long lived
- ☐ Making large, infrequent commits



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✓ **Correct** 1/1 Points

9. A critical security vulnerability is discovered in the production system. According to best practices, from which branch should a hotfix branch be created? \*

- ☐ Develop branch
- ☐ Feature branch
- ☐ Release branch
- ☒ Main branch
- ☐ Any branch containing the bug



✓ **Correct** 1/1 Points

10. After completing a hotfix and merging it into the main branch, what is the NEXT best practice step? \*

- ☐ Leave the hotfix branch for future fixes
- ☐ Rebase the main branch onto the hotfix branch



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✓ **Correct** 1/1 Points

7. Which practice MOST effectively reduces merge conflicts when working with feature branches? \*

- ☐ Keeping feature branches long lived
- ☐ Making large, infrequent commits
- ☒ Merging changes early and frequently ✓
- ☐ Avoiding pull requests
- ☐ Working directly on the main branch

✓ **Correct** 1/1 Points

8. Why is GitFlow often considered unsuitable for small student projects or small teams? \*

- ☐ It does not support feature branches
- ☒ It enforces a complex branching model that may be unnecessary ✓



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4. A team uses multiple developers, continuous deployment, and frequent releases. Which of the following is NOT considered a recognized Git workflow or branching strategy? \*

☐ GitFlow

☐ GitHub Flow

☐ GitLab Flow

☒ Monolithic Flow ✓

☐ Trunk-Based Development

✓ **Correct** 1/1 Points

5. What distinguishes GitHub Flow from GitFlow in terms of branch structure? \*

☐ GitHub Flow relies on long lived develop and release branches

☐ GitHub Flow prevents direct collaboration between developers





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- ☐ Squash all previous commits into one commit
- ☒ Revert the commit on main and move the work into a separate feature branch ✓

✓ **Correct** 1/1 Points

12. Which command creates and switches to a new branch in Git? \*

- ☐ git branch <branch\_name>
- ☐ git checkout <branch\_name>
- ☒ git checkout -b <branch\_name> ✓
- ☐ git merge <branch\_name>
- ☐ git switch <branch\_name>

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✓ **Correct** 1/1 Points

3. A developer is halfway through implementing Feature A when an urgent production bug is reported. Which Git feature MOST directly enables the developer to address the bug without losing progress on Feature A? \*

- ☐ Commit squashing
- ☐ Rebasing
- ☒ Branch switching ✓
- ☐ Tagging releases
- ☐ Cloning the repository again

✓ **Correct** 1/1 Points

4. A team uses multiple developers, continuous deployment, and frequent releases. Which of the following is NOT considered a recognized Git workflow or branching strategy? \*



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✓ **Correct** 1/1 Points

10. After completing a hotfix and merging it into the main branch, what is the NEXT best practice step? \*

- ☐ Leave the hotfix branch for future fixes
- ☐ Rebase the main branch onto the hotfix branch
- ☒ Merge the hotfix changes back into the development branch and delete the hotfix branch ✓
- ☐ Convert the hotfix branch into a feature branch
- ☐ Undo all previous commits

✓ **Correct** 1/1 Points

11. A developer accidentally commits unfinished code directly to the main branch. The code has already been pushed to the remote repository, but no release has been made yet. Which is the MOST appropriate action based on Git



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✓ **Correct** 1/1 Points

5. What distinguishes GitHub Flow from GitFlow in terms of branch structure? \*

- ☐ GitHub Flow relies on long lived develop and release branches
- ☐ GitHub Flow prevents direct collaboration between developers
- ☒ GitHub Flow uses a single long lived main branch with short lived feature branches ✓
- ☐ GitHub Flow is designed exclusively for enterprise scale projects
- ☐ GitHub Flow does not require merging branches

✓ **Correct** 1/1 Points

6. A Pull Request (PR) is created from a feature branch to the main branch. Which is the PRIMARY reason for this step in GitHub Flow? \*

- ☐ To archive the feature branch before deletion





✓ Correct 1/1 Points

1. Which of the following is the **BEST** description of Version Control? \*

- ☐ A system for automatically backing up files to the cloud.
- ☒ A system that records changes to a file or set of files over time, allowing you to recall specific versions later. ✓
- ☐ A method for organizing files into folders and subfolders.
- ☐ A tool for compressing files to save disk space.

✓ Correct 1/1 Points

2. Which of the following is **NOT** a type of Version Control System ? \*

- ☐ Local Version Control Systems
- ☐ Centralized Version Control Systems
- ☒ Manual version control Systems ✓
- ☐ Distributed Version Control Systems



✓ Correct 1/1 Points

9. Which of the following services is NOT a Git repository hosting service? \*

- ☐ GitHub
- ☐ GitLab
- ☐ Bitbucket
- ☒ Stack Overflow



✗ Incorrect 0/1 Points

10. Which of the following BEST explains why Git is considered the *de-facto standard* for version control systems? \*

- ☒ Git was officially declared as the standard by an international organization
- ☐ Git is the oldest version control system available
- ☐ Git is widely adopted by the software industry and open-source community without formal standardization
- ☐ Git can only be used with GitHub







✓ Correct 1/1 Points

5. What is the purpose of a Remote Repository in Git? \*

- ☐ To delete local repositories
- ☐ To act as a backup without sharing with others
- ☒ To store a centralized version of the project for collaboration ✓
- ☐ To rename files automatically

✓ Correct 1/1 Points

6. What is the primary difference between Git and GitHub? \*

- ☒ Git is a version control system, while GitHub is a cloud-based repository hosting service ✓
- ☐ GitHub is used for programming, while Git is used for designing
- ☐ Git is a centralized version control system, while GitHub is a distributed system
- ☐ GitHub replaces Git entirely

✓ Correct 1/1 Points



✓ Correct 1/1 Points

**3. Which of the following is a characteristic of Centralized Version Control Systems ? \***

- ☐ Each user maintains a complete local copy of the repository.
- ☐ They follow a peer-to-peer (P2P) approach.
- ☐ They are primarily used in open-source projects.
- ☒ They require a central authority to maintain the central repository. ✓

✓ Correct 1/1 Points

**4. Which of the following BEST explains why Distributed Version Control Systems (DVCS) are suitable for open-source projects? \***

- ☐ They require a central server to approve all changes
- ☒ Each contributor has a complete copy of the repository, enabling independent work ✓
- ☐ They only allow one developer to modify files at a time
- ☐ They prevent contributors from working offline

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7. Which feature is available in GitHub but NOT in Git itself? \*

- ☐ Branching
- ☐ Commit tracking
- ☒ Code hosting and collaboration ✓
- ☐ Merging

✓ Correct 1/1 Points

8. In Git, what is a "commit"? \*

- ☐ A request to merge changes from one branch to another.
- ☒ A snapshot of changes in the local repository at a specific point in time. ✓
- ☐ A command to download the latest version of a remote repository.
- ☐ A tool for resolving conflicts between different versions of a file.

✓ Correct 1/1 Points

9. Which of the following services is NOT a Git repository hosting service? \*

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✓ **Correct** 1/1 Points

7. Which of the following is an architectural pattern commonly supported by software frameworks to promote separation of concerns? \*

☐ FIFO

☒ MVC



☐ LIFO

☐ Bubble Sort

✓ **Correct** 1/1 Points

8. What does Inversion of Control (IoC) mean in the context of frameworks? \*

☐ The developer manually controls object creation and dependencies.

☐ It is a database optimization technique used to manage query execution.

The framework controls the flow of execu-



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✓ **Correct** 1/1 Points

3. From a software engineering perspective, why is repeatedly rewriting generic functionalities for each new application considered a poor practice? \*

- ☐ It reduces developer creativity
- ☐ It increases system flexibility
- ☒ It leads to duplicated effort, higher maintenance cost, and increased risk of errors ✓
- ☐ It violates programming language syntax rules

✓ **Correct** 1/1 Points

4. What is the primary purpose of using a software framework in application development? \*

- ☐ It eliminates the need for coding entirely
- ☒ It provides a structured foundation for software development, reducing redundant work ✓



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✓ **Correct** 1/1 Points

6. Select the false statement about Software Frameworks. \*

- ☐ Frameworks enforce best practices such as coding standards, directory structures, and naming conventions.
- ☒ A software framework allows developers to directly modify its core source code as needed. ✓
- ☐ Using a framework can help speed up development by providing reusable components for generic functionalities.
- ☐ Frameworks often provide tooling support such as dependency management and boilerplate code generation.

✓ **Correct** 1/1 Points

7. Which of the following is an architectural pattern commonly supported by software frameworks to promote separation of concerns? \*





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✓ **Correct** 1/1 Points

9. Which of the following functionalities is LEAST likely to be directly provided by a typical web application framework and usually requires external tools or services? \*

- ☐ User authentication
- ☐ Database connectivity
- ☐ Routing
- ☒ Application monitoring and debugging intelligence ✓

✓ **Correct** 1/1 Points

10. Choosing an inappropriate software framework can negatively impact user experience, especially when the framework includes many unused features. What is the MOST likely reason for this impact? \*

- ☐ Frameworks inherently slow down all applications regardless of usage

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✓ **Correct** 1/1 Points

4. What is the primary purpose of using a software framework in application development? \*

- ☐ It eliminates the need for coding entirely
- ☒ It provides a structured foundation for software development, reducing redundant work ✓
- ☐ It allows developers to modify core framework code freely
- ☐ It is only useful for small-scale applications.

✓ **Correct** 1/1 Points

5. Which statement MOST accurately distinguishes a software framework from a library in terms of control flow and application structure? \*

- ☐ A framework provides reusable functions, while a library enforces architectural patterns.
- ☐ A framework controls the overall application



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✓ **Correct** 1/1 Points

2. Which of the following activities is LEAST associated with the software engineering process, even though it may occur during software development? \*

- ☐ Analyzing user requirements and designing solutions
- ☐ Selecting architectural patterns and technologies
- ☒ Implementing a single algorithm to solve a specific problem ✓
- ☐ Deploying, releasing, and maintaining software

✓ **Correct** 1/1 Points

3. From a software engineering perspective, why is repeatedly rewriting generic functionalities for each new application considered a poor practice? \*

- ☐ It reduces developer creativity



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✓ **Correct** 1/1 Points

8. What does Inversion of Control (IoC) mean in the context of frameworks? \*

- ☐ The developer manually controls object creation and dependencies.
- ☐ It is a database optimization technique used to manage query execution.
- ☒ The framework controls the flow of execution instead of the programmer. ✓
- ☐ It is a method for improving UI performance.

✓ **Correct** 1/1 Points

9. Which of the following functionalities is LEAST likely to be directly provided by a typical web application framework and usually requires external tools or services? \*

- ☐ User authentication
- ☐ Database connectivity



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Points: 10/10

✓ **Correct** 1/1 Points

1. Which statement BEST differentiates Programming from Software Engineering?

\*

- ☐ Programming focuses on tools while software engineering focuses on frameworks
- ☒ Programming is a subset of software engineering ✓
- ☐ Software engineering only applies to large projects
- ☐ Programming excludes testing and debugging

✓ **Correct** 1/1 Points

2. Which of the following activities is LEAST associated with the software engineering process, even though it may occur during





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✓ **Correct** 1/1 Points

10. Choosing an inappropriate software framework can negatively impact user experience, especially when the framework includes many unused features. What is the MOST likely reason for this impact? \*

- ☐ Frameworks inherently slow down all applications regardless of usage
- ☒ Framework constraints and unused components can affect performance, scalability, and maintainability ✓
- ☐ Frameworks restrict user interface design choices
- ☐ Frameworks prevent effective debugging and testing

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✓ **Correct** 1/1 Points

5. Which statement MOST accurately distinguishes a software framework from a library in terms of control flow and application structure? \*

- ☐ A framework provides reusable functions, while a library enforces architectural patterns.
- ☒ A framework controls the overall application flow and calls user code, whereas a library is called by user code when needed. ✓
- ☐ A library defines the lifecycle of an application, while a framework only provides helper utilities.
- ☐ A framework and a library differ only in size and number of features.

✓ **Correct** 1/1 Points

6. Select the false statement about Software Frameworks. \*

- ☐ Frameworks enforce best practices such as coding standards, directory structures, and

